



STATISTICAL ANALYSIS OF FIRST-MOVE STRATEGY AND GAME DYNAMICS IN 11x11 HEX

“*Shall we play a game?*” Joshua, the supercomputer, 1983 movie *War Games*

ABSTRACT

Hex is a two-player strategy game where the players' goal is forming an unbroken path across a board of hexagonal spaces. Danish mathematician Piet Hein invented the game in 1942, and subsequently, American mathematician John Nash introduced it at Princeton in 1948. Nash also proved that the first player has a winning strategy, but how that plays out in real games is less clear. This project examines 11x11 Hex games to analyze first-move selection, center versus edge openings, and game duration. We analyze over 2,000 games, with 118 unique starting moves, using manually collected data from games played on PlayHex.org. We apply chi-square tests to assess whether move distributions deviate from randomness, ANOVA and t-tests to analyze the relationships between first moves, game lengths, and win rates, and proportion tests to evaluate win rates by starting move. We drew on the theoretical path framework developed by Cameron Browne, whose book *Hex Strategy: Making the Right Connections* formalized many core ideas about move strength and group connectivity. Our findings provide support to existing mathematical principles underlying optimal Hex gameplay.

INITIAL RESEARCH QUESTIONS

- Are certain first moves more frequent in wins?
- Do opening locations affect win rates and game length?
- Does the swap rule affect win probability?
- Do optimal moves favor unbroken or near-unbroken chains?

HYPOTHESES

- Winning games will have non-random move distributions with key first move positions appearing more frequently.
- Moves that align with Cameron Browne's strategies will be more common in winning games.
- Strategic first moves will correlate with certain game durations.

RESEARCH AND STATISTICAL METHODS

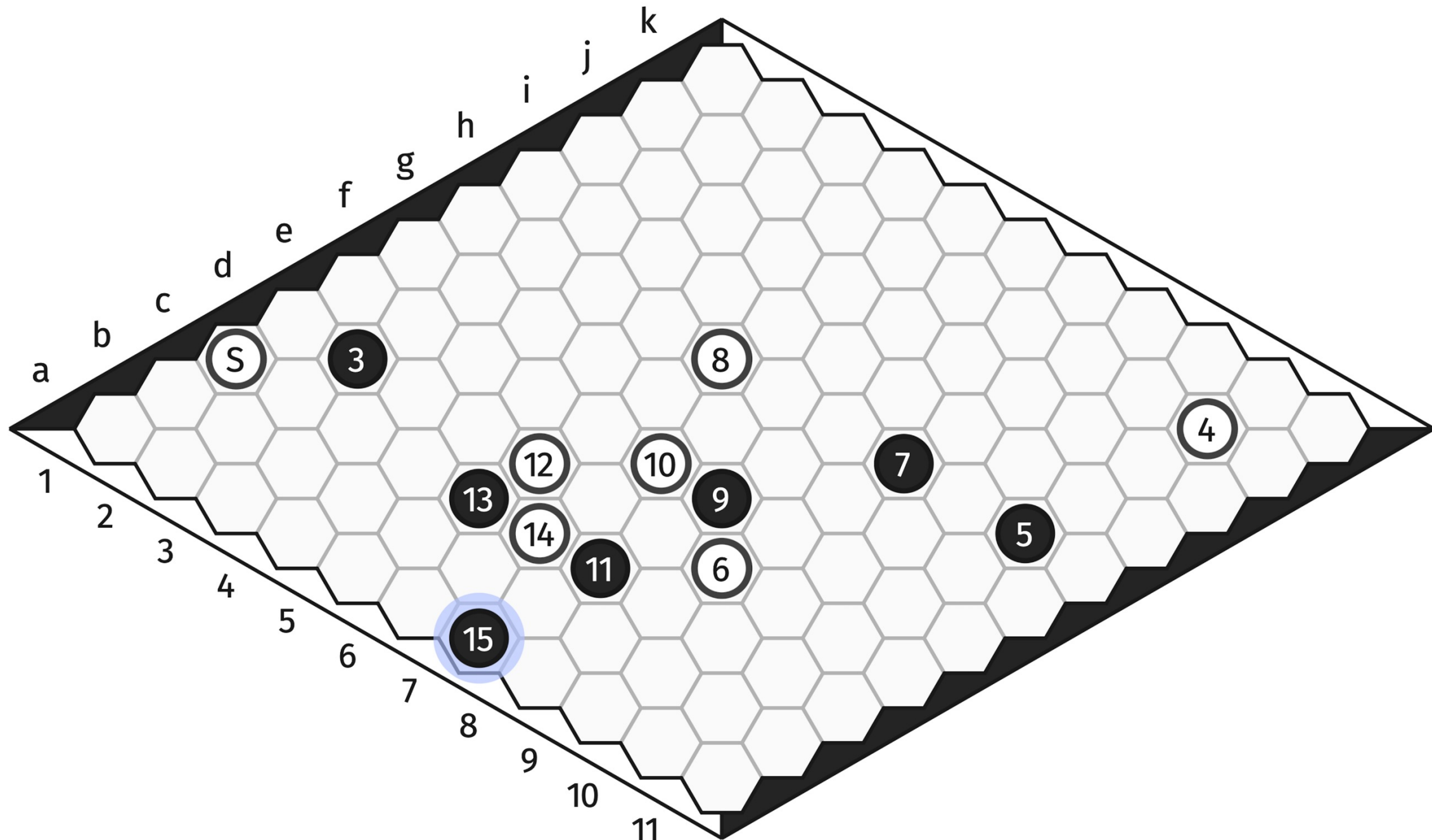
Theories Referenced	Nash's first-player winning strategy (1949, non-constructive proof) Cameron Browne's Hex Strategy (path theory, adjacency, and chain logic)
Tests Used	Monte Carlo Chi-Square, T-Test, ANOVA, Proportion Tests, Fisher Test

DATA COLLECTION AND CLEANING

Raw Data Acquisition	Manually acquired JSON files from over 4,000 1v1 games played on PlayHex.Org via Chrome DevTools
CSV Conversion	Extracted select metadata: board size, outcome, winner, timestamps, move history. Omitted player names and ranking for focus on gameplay structure
Data Cleaning	Removed games with pass moves (@0), early quits (≤5 moves), and non-11x11 boards, standardized outcomes, converted time fields to datetime, calculated game durations

EXPLORATORY DATA ANALYSIS

Tools Used	R programming language (dplyr, ggplot2, tidy, etc). Prolog programs referenced in order to simulate similar style path logic translated. <i>Hex Strategy: Making the Right Connections</i> by Cameron Browne
Move Categorization	First moves classified into Center, Edge., and Corner based on modified Browne framework
Data Analysis	Analyzed move frequency, win rate by move, game length distribution, and top first moves; Prepared inputs for statistical testing



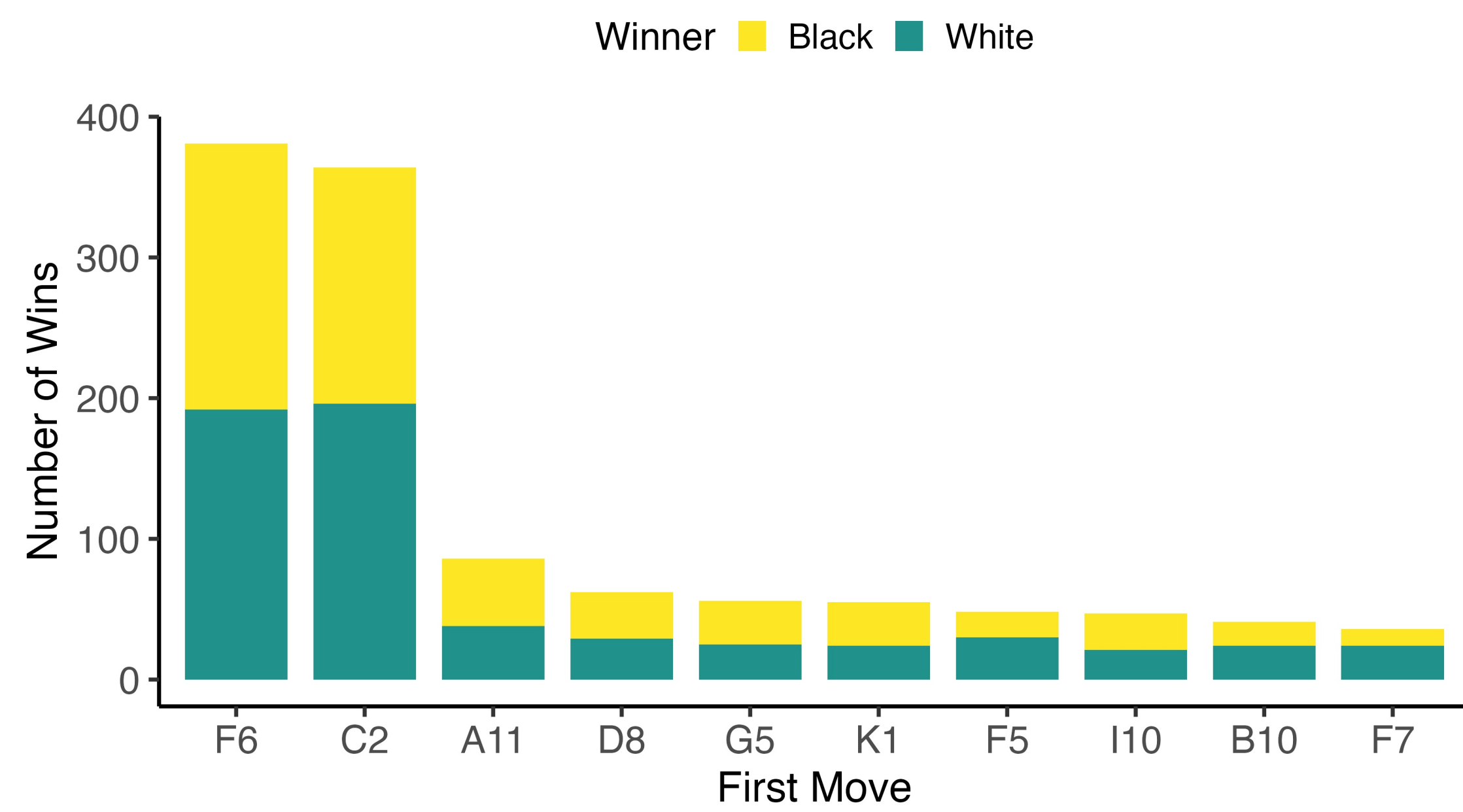
This Hex board shows a game in progress with labeled move numbers. The board uses a coordinate system where columns are labeled a–k and rows 1–11. The first move, marked S, was placed by Black at D1. White chose to swap, claiming the first move and now playing as Black. The most recent move, 15, was placed by Black at A8, highlighted with a blue circle.

EDA RESULTS

- 2,414 total games analyzed
- 1,076 completed, 1,107 resigned, 231 flagged
- All completed games contained unbroken winning chains
- Resigned/flagged games often had near-unbroken chains (≤2 moves from win)
- Most first moves occurred outside strategic zones (center, edge, corner)
- “Center” moves were more common in high-frequency openings

BRIEF LOOK AT FIRST MOVES

Win Counts by First Move (Top 10)

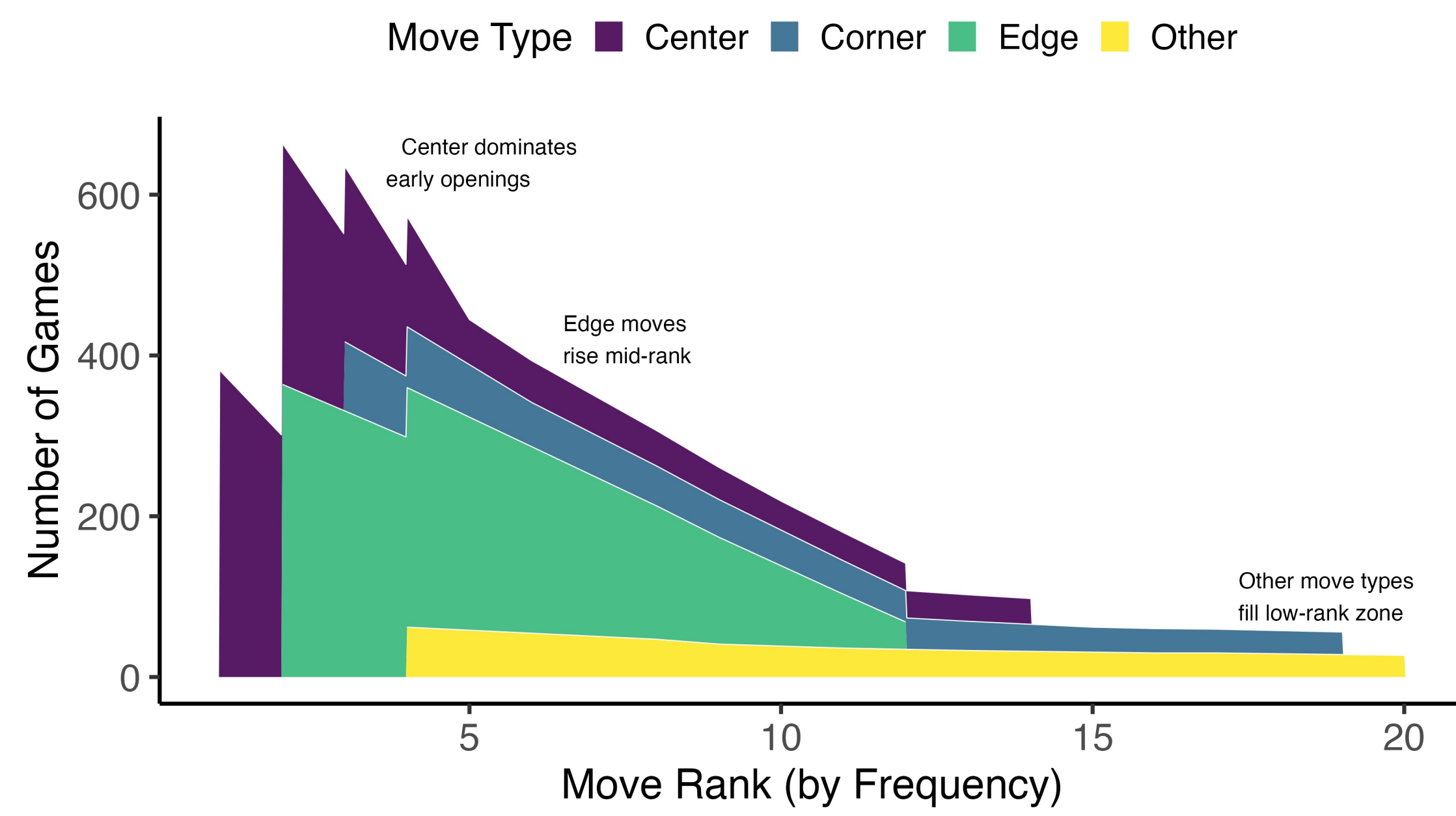


A few opening moves are used far more frequently than others. F6, C2, and A11 dominate early game strategies.

MOVE CLASSIFICATION

Category	Description	Valid Moves
Center	Associated with strategic path options and higher connectivity.	E5, E6, E7, F5, F6, F7, G5, G6, G7
Edge	Adjacent to the edge of board	A3, A9, K3, K9, C, J2, C10, J10
Corner	Extreme corners of the board.	A1, K1, A11, K11

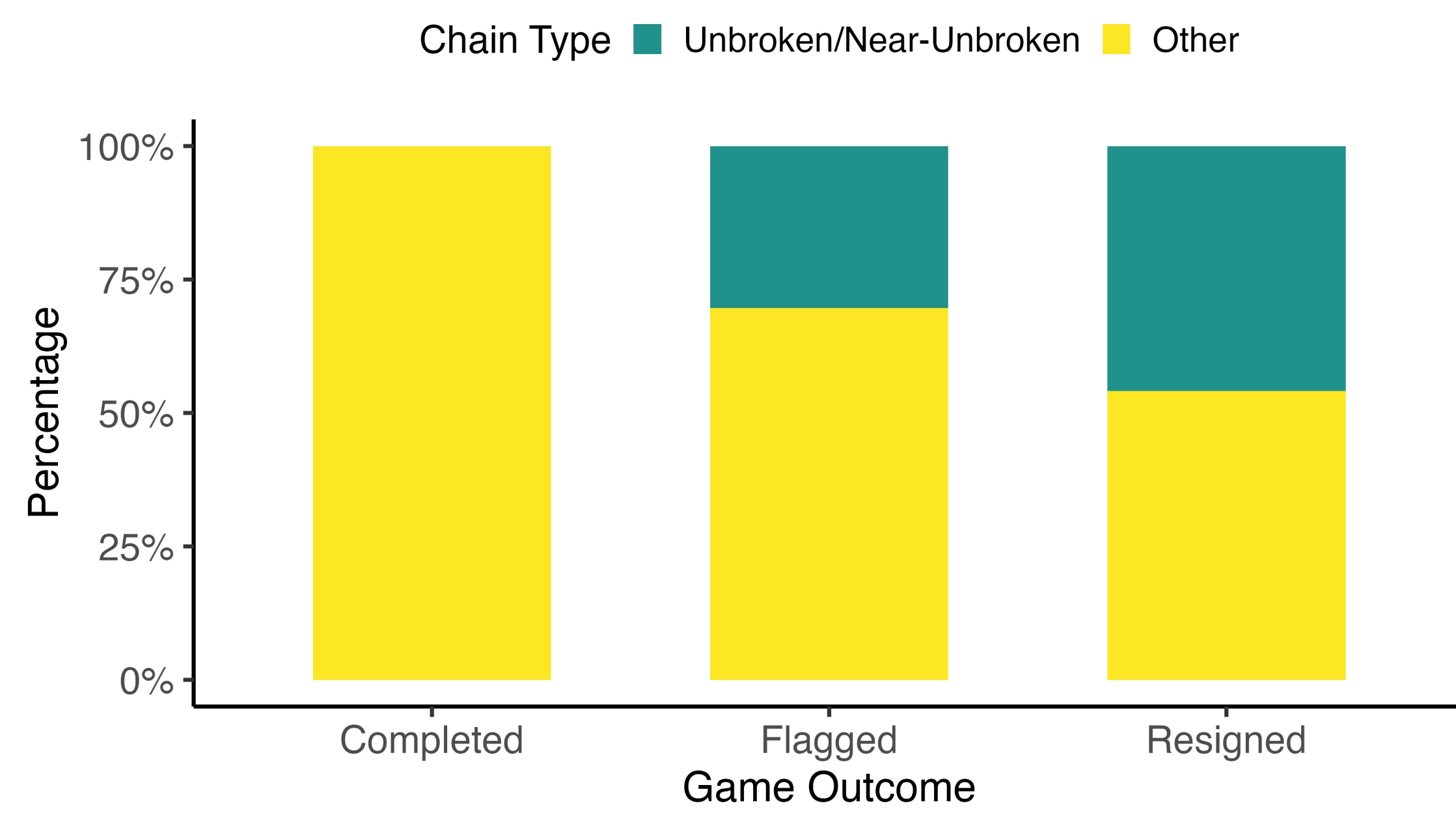
Top 20 First Moves by Type



Most common first moves are in the center or edge of the board, with corners rarely chosen as openings.

PATHS IN HEX – HOW CLOSE TO A WIN?

Chain Types by Game Outcome

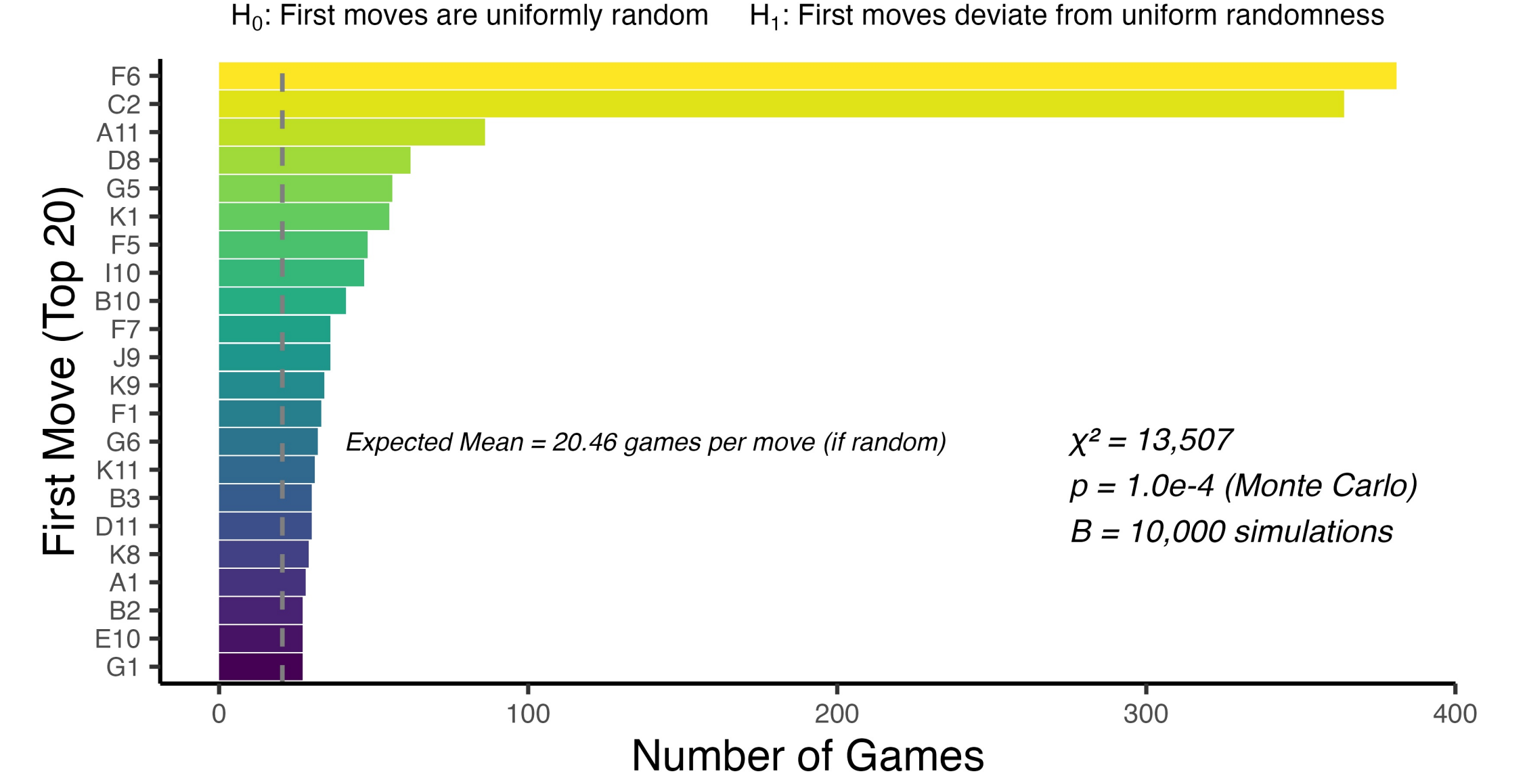


We analyzed games that ended in resignation or time expiration to check if a player was close to forming a winning path. This was particularly important for identifying near-complete game strategy.

STATISTICAL ANALYSIS

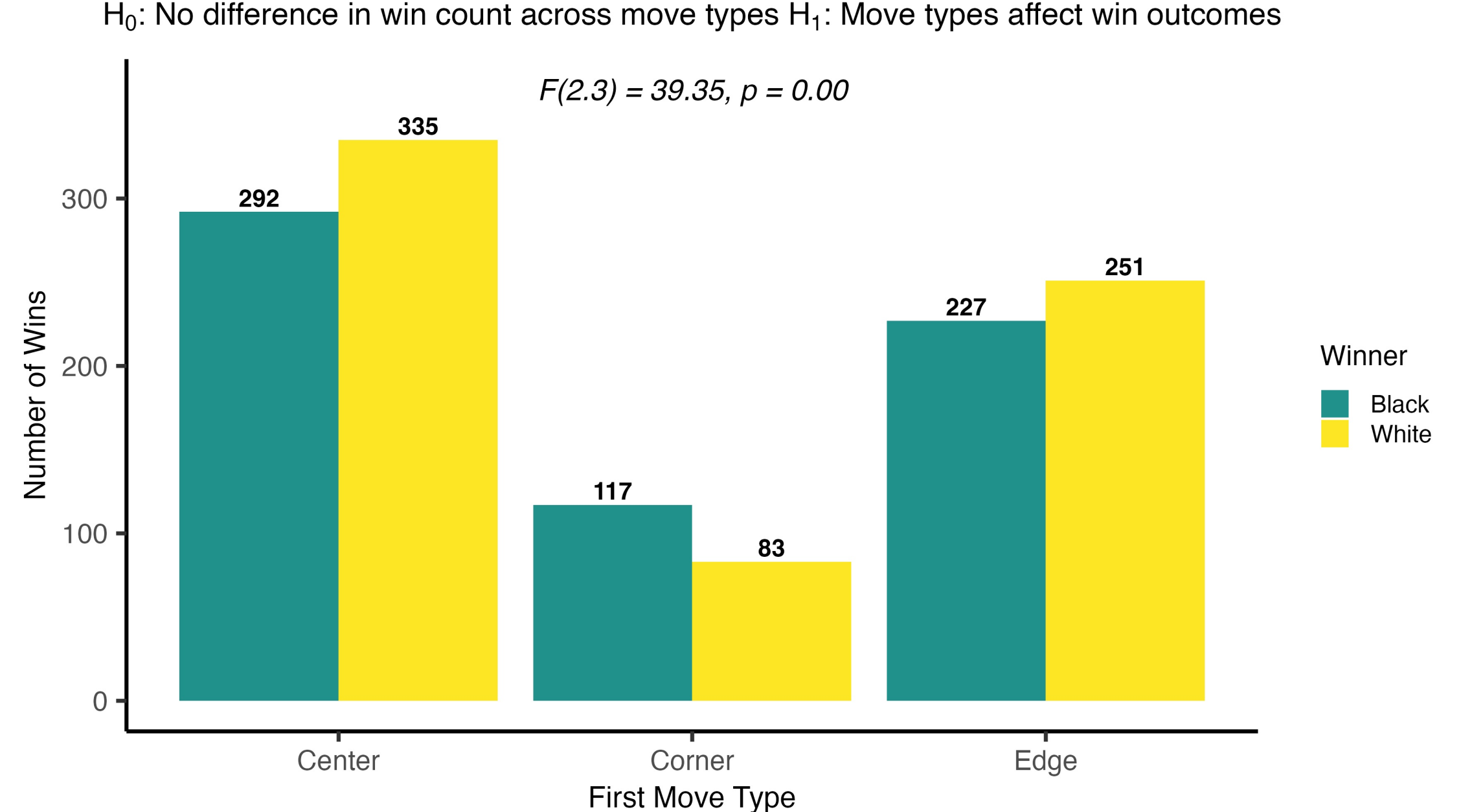
Question	Test	P-Value	Significant?
Are first moves random?	Monte Carlo Chi-Square	0.0001	Yes
Win rate: Edge vs. Center	T-Test	0.1255	No
Win rate: Center/Edge/Corner	ANOVA	0.007	Yes
Win rate: Top 3 moves vs. others	Proportion Test	0.6571	No
First move affects game length?	ANOVA	0.0004	Yes
Swap move affects win rate?	Fisher test	0.929	No
Top 3 moves have more near-unbroken chains?	Proportion	< 2.2e-16	Yes
Categorized moves have more unbroken and near-unbroken chains?	Proportion	< 2.2e-16	Yes

Chi-square Test: Are first moves random?



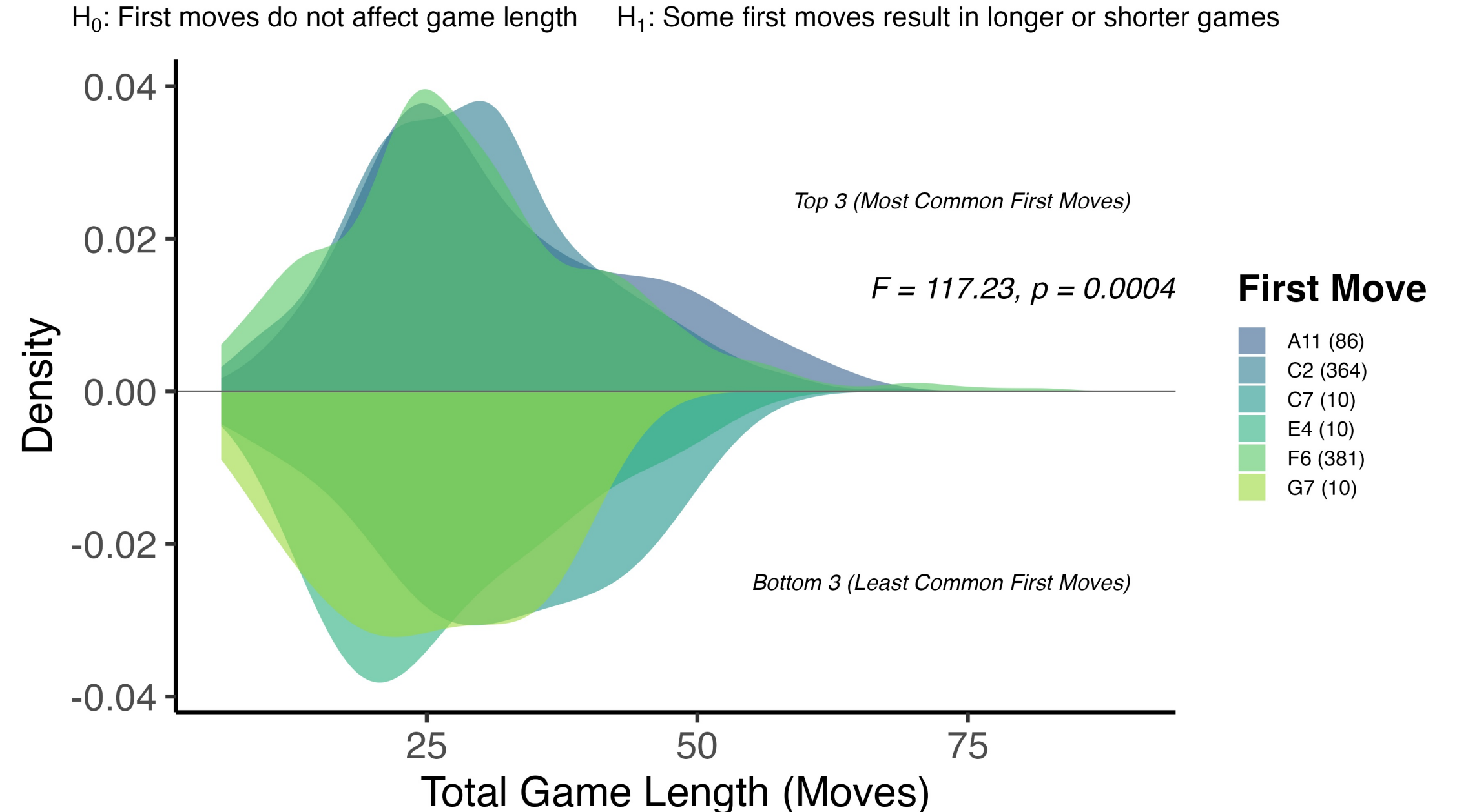
Monte- Carlo permutations were added due to sample size of first moves. $\chi^2 = 13,507$, $p < 0.0001 \rightarrow$ We reject the null hypothesis. Conclusion: First moves are not uniformly distributed across the board.

ANOVA Test: Win Distribution by First Move Type



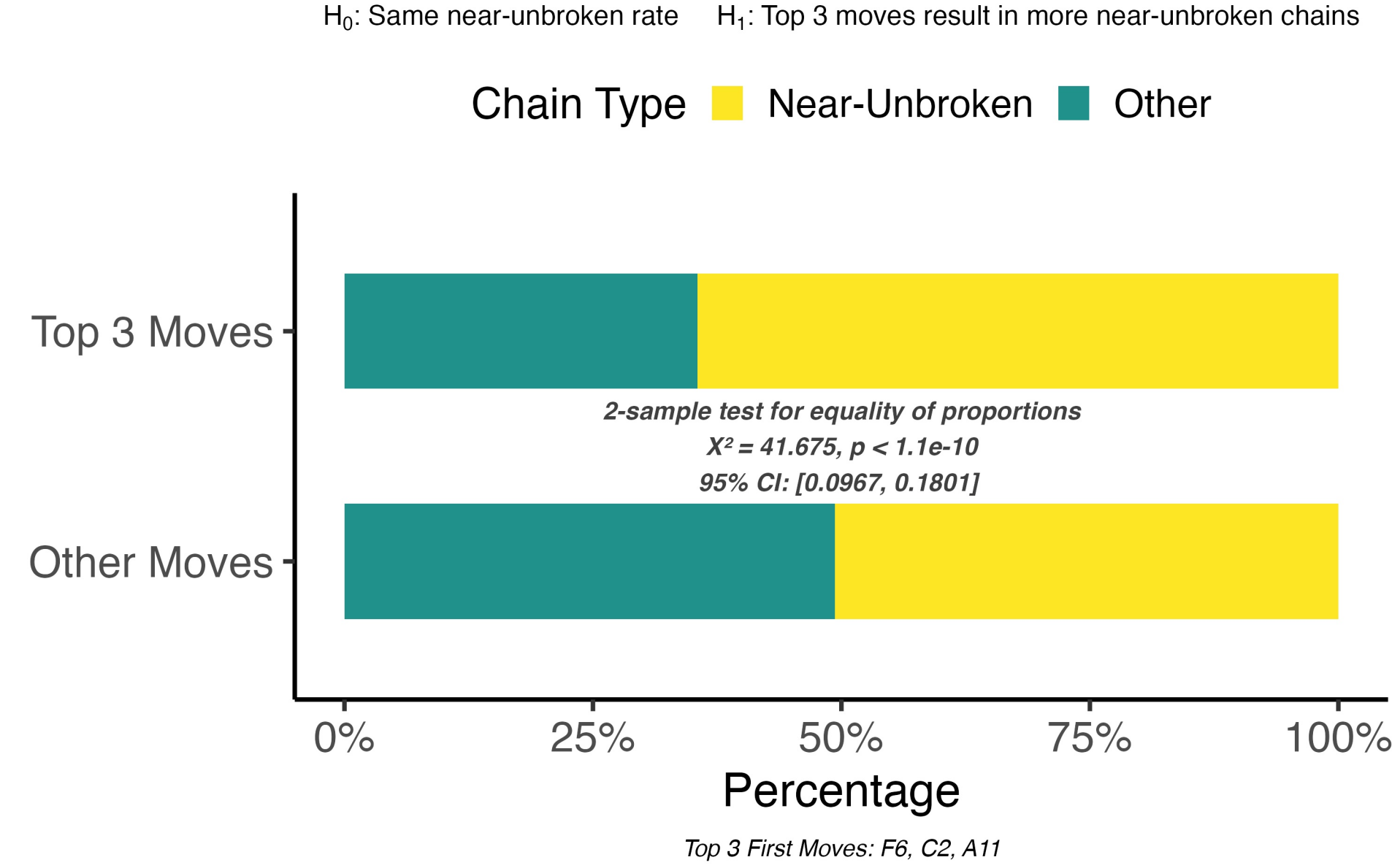
$F(2, 3) = 39.35$, $p = 0.007 \rightarrow$ We reject the null hypothesis. Conclusion: Win rates differ significantly by move type.

ANOVA Test: First Move Affects Game Length



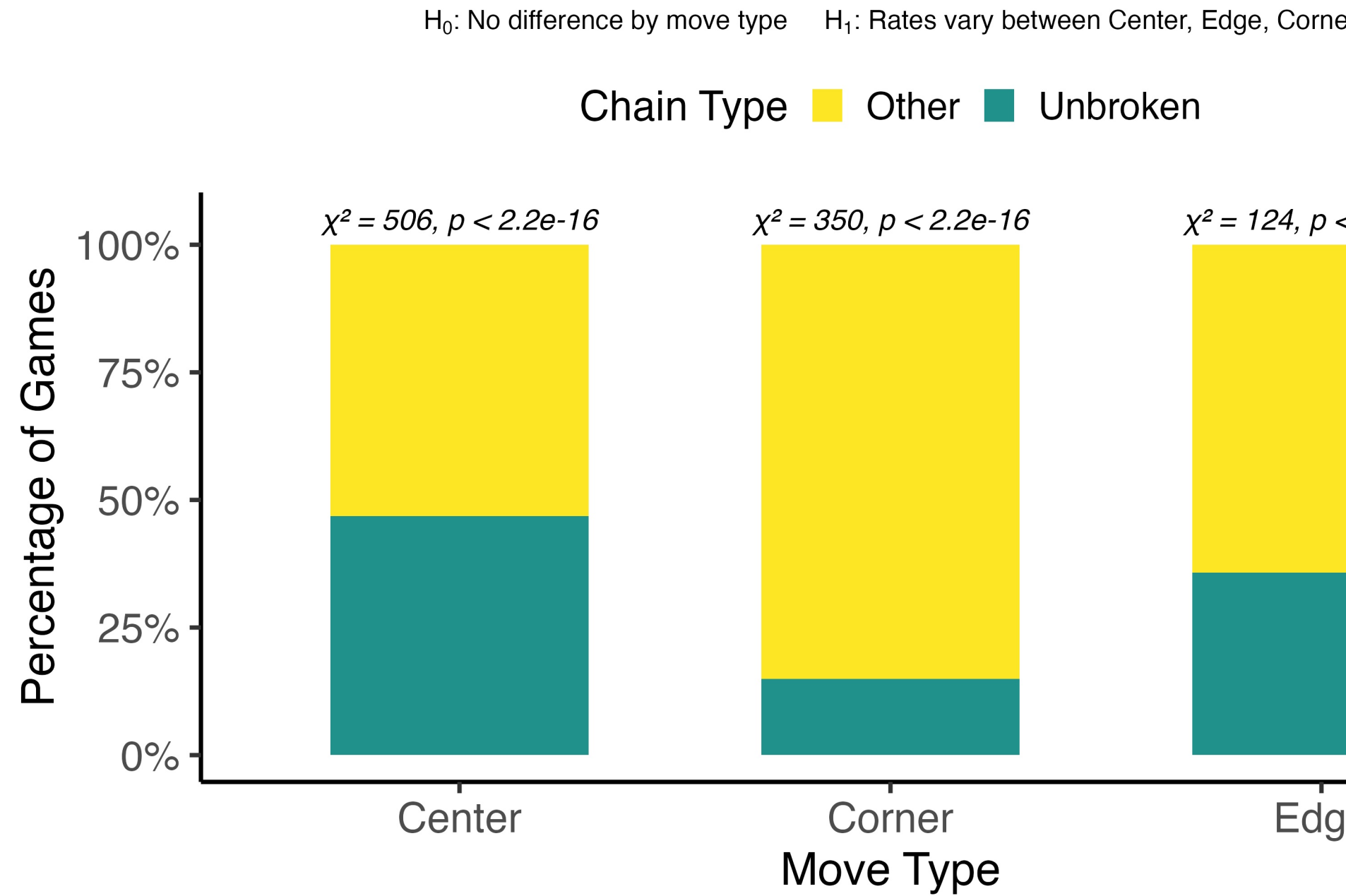
$F(117, 2296) = 1.52$, $p = 0.0004 \rightarrow$ We reject the null hypothesis. Conclusion: First move influences game length.

Proportion Test: Top 3 First Moves Have More Near-Unbroken Chains



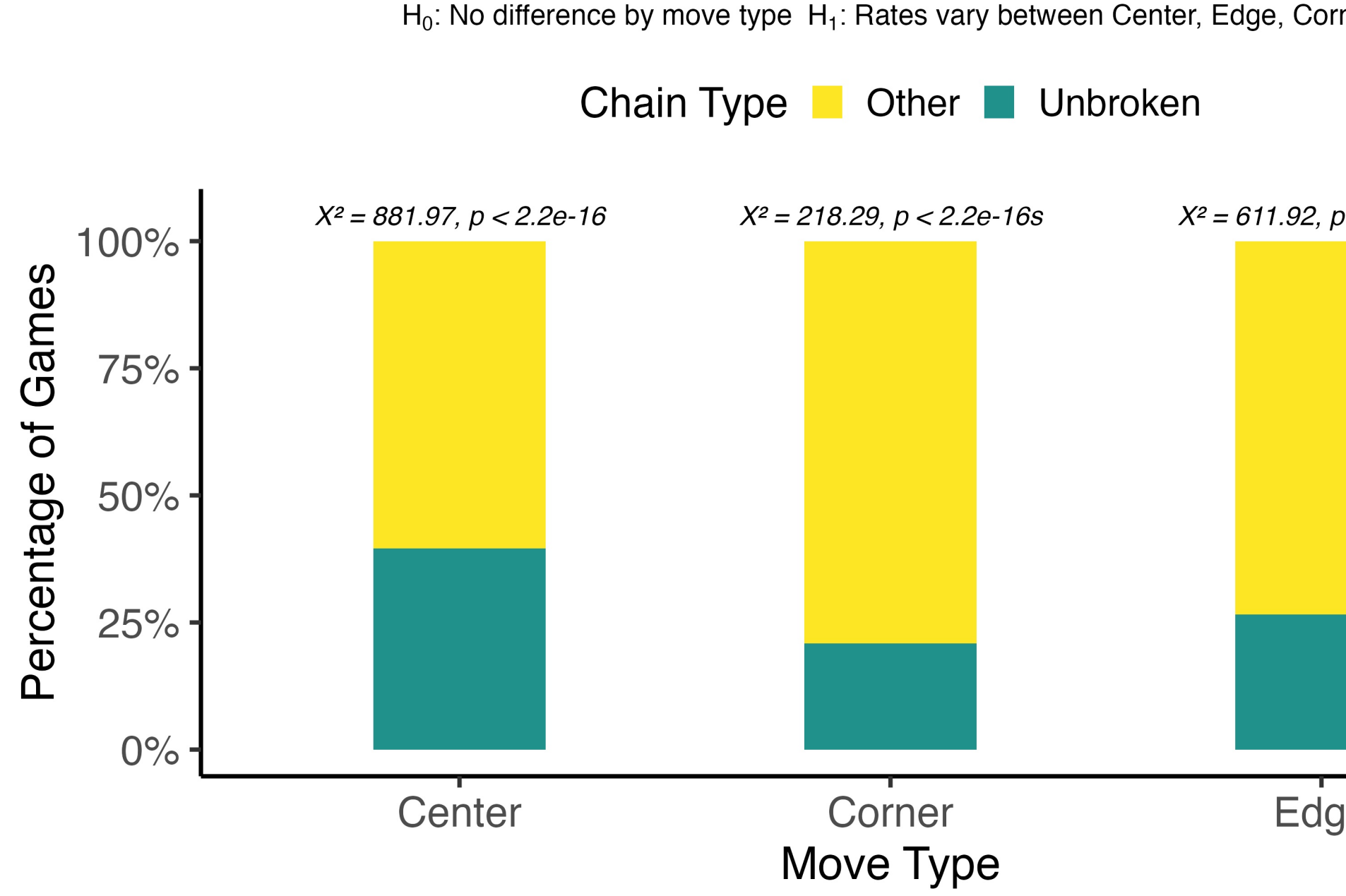
$\chi^2 = 41.68$, $p < 0.0000000001 \rightarrow$ We reject the null hypothesis. Conclusion: Near-unbroken chains occur more often with top 3 moves.

Proportion Test on Near-Unbroken Chains



$\chi^2 = 41.68$, $p < 0.0000000001 \rightarrow$ We reject the null hypothesis. Conclusion: Near-unbroken chain rates differ by move type.

Proportion Test on Unbroken Chains



Center: $\chi^2 = 882$, Edge: $\chi^2 = 612$, Corner: $\chi^2 = 218 \rightarrow$ All $p < 0.0001 \rightarrow$ We reject the null hypothesis. Conclusion: Unbroken chain rates differ by move type.

CONCLUSION

Our statistical analysis revealed that the Chi-square tests showed a strong concentration of opening choices like F6 and C2. The ANOVA statistical test confirmed that move type (center, edge, corner) influences both win rate and game length. Top first moves were significantly associated with forming unbroken or nearly unbroken chains in games that ended in resign or flag outcome status, which supported the theoretical expectations from Browne's path model. The swap rule showed no significant impact on game outcomes and that the most popular openings did not always result in a higher win rate. These results suggest that strategic depth goes beyond the first move alone and further game play tactics are required to achieve a higher win rate.

FUTURE RESEARCH

Further research could deepen the application of Cameron Browne's theories by implementing additional strategic elements which he discusses, such as ladders, bridges, and virtual connections. Such an implementation would require more advanced pathing models and tactical programming logic, which potentially integrate rule-based AI or graph-theoretical approaches – and a switch from R programming to Python with possibly using pyswip library, which is a SWI-Prolog Python interface.